

Table 1: NanoAOD 2023BPix datasets used in the analysis

Sample	NanoAOD dataset
Data (4 samples)	
Muon0_Run2023D_v1	/Muon0/Run2023D-22Sep2023_v1-v1/NANOAOD
Muon0_Run2023D_v2	/Muon0/Run2023D-22Sep2023_v2-v1/NANOAOD
Muon1_Run2023D_v1	/Muon1/Run2023D-22Sep2023_v1-v1/NANOAOD
Muon1_Run2023D_v2	/Muon1/Run2023D-22Sep2023_v2-v1/NANOAOD
Simulation (106 samples)	
GluGluHto2Mu_M120	/GluGluHto2Mu_M-120_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
GluGluHto2Mu	/GluGluHto2Mu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
GluGluHto2Mu_amcatnlo	/GluGluHto2Mu_M-125_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
GluGluHto2Mu_MinNLO	/GluGluHto2Mu_M-125_TuneCP5_13p6TeV_powhegMinNLO-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
GluGluHto2Mu_M-125_TuneCP5Down_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM	
GluGluHto2Mu_tuneUp	/GluGluHto2Mu_M-125_TuneCP5Up_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
GluGluHto2Mu_M130	/GluGluHto2Mu_M-130_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
VBFHto2Mu_M125_powheg	/VBFHto2Mu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
VBFHto2Mu_m125_tuneCP5Up_amcatnlo	/VBFH_Hto2Mu_M-125_TuneCP5Up_13p6TeV_amcatnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
VBFHto2Mu_m125_tuneCP5Down_amcatnlo	/VBFH_Hto2Mu_M-125_TuneCP5Down_13p6TeV_amcatnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
VBFHto2Mu_M125_amcatnlo	/VBFH_Hto2Mu_M-125_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
ZH_Hto2Mu	/ZH_Hto2Mu_ZtoAll_M-125_TuneCP5_13p6TeV_powheg-minlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
WplusH_Hto2Mu	/WplusH_Hto2Mu_WtoAll_M-125_TuneCP5_13p6TeV_powheg-minlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
WminusH_Hto2Mu	/WminusH_Hto2Mu_WtoAll_M-125_TuneCP5_13p6TeV_powheg-minlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
TTH_Hto2Mu	/TTH_Hto2Mu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
VBFHto2Mu_M120_amcatnlo	/VBFH_Hto2Mu_M-120_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
VBFHto2Mu_M130_amcatnlo	/VBFH_Hto2Mu_M-130_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
DYto2Mu_MLL_105to160_amcatnloFXFX	/DYto2Mu-2Jets_MLL-105To160_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
DYto2L_M_10to50_amcatnloFXFX	/DYto2L-2Jets_MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v4/NANOAODSIM
	/DYto2L-2Jets_MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2_ext1-v4/NANOAODSIM
DYto2L_M_50_0J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_0J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v4/NANOAODSIM
DYto2L_M_50_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v4/NANOAODSIM
DYto2L_M_50_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v5/NANOAODSIM
DYto2L_M_50_amcatnloFXFX	/DYto2L-2Jets_MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v4/NANOAODSIM
DYto2L_M_50_PTLT_40to100_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLT-40to100_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v1/NANOAODSIM
DYto2L_M_50_PTLT_40to100_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLT-40to100_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v1/NANOAODSIM
DYto2L_M_50_PTLT_100to200_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLT-100to200_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v1/NANOAODSIM
DYto2L_M_50_PTLT_100to200_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLT-100to200_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v1/NANOAODSIM
DYto2L_M_50_PTLT_200to400_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLT-200to400_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
	/DYto2L-2Jets_MLL-50_PTLT-200to400_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6_ext1-v2/NANOAODSIM
DYto2L_M_50_PTLT_200to400_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLT-200to400_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
	/DYto2L-2Jets_MLL-50_PTLT-200to400_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6_ext1-v2/NANOAODSIM
DYto2L_M_50_PTLT_400to600_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLT-400to600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v3/NANOAODSIM
	/DYto2L-2Jets_MLL-50_PTLT-400to600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6_ext1-v3/NANOAODSIM
DYto2L_M_50_PTLT_400to600_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLT-400to600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
	/DYto2L-2Jets_MLL-50_PTLT-400to600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6_ext1-v2/NANOAODSIM
DYto2L_M_50_PTLT_600_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLT-600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
	/DYto2L-2Jets_MLL-50_PTLT-600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6_ext1-v2/NANOAODSIM
DYto2L_M_50_PTLT_600_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTLT-600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
	/DYto2L-2Jets_MLL-50_PTLT-600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6_ext1-v2/NANOAODSIM
EWK_2L2J_madgraph_herwig	/EWK_2L2J_TuneCH3_13p6TeV_madgraph_herwig7/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
EWK_2Mu2J_MLL_105to160_herwig	/EWK_2Mu2J_Bin-MLL-105to160_TuneCH3_13p6TeV_madgraph_herwig7/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAODSIM
EWK_2Mu2J_MLL_105to160_pythia	/EWK-2Mu2J_M2Mu-105to160_M2J-120_TuneCP5_13p6TeV_madgraph-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v3/NANOAODSIM
ggZH_Hto2B_Zto2Q	/ggZH_Hto2B_Zto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
ggZH_Hto2B_Zto2L	/ggZH_Hto2B_Zto2L_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAODSIM
GluGluHto2B_M125	/GluGluHto2B_M-125_TuneCP5_13p6TeV_powheg-minlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
GluGluHto2Tau_M125	/GluGluHto2TauTau_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
GluGluHto2Tau_UnCorrelatedDecay_Filtered	/GluGluHto2TauUnCorrelatedDecay_Filtered_M-125_CP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM
GluGluHto2Tau_UnCorrelatedDecay_UnFiltered	/GluGluHto2TauUnCorrelatedDecay_M-125_CP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAODSIM

Continued on next page

Table 1: NanoAOD 2023BPix datasets used in the analysis (continued)

Sample	NanoAOD dataset
GluGluHto2Wto2L2Nu_M125	/GluGluHto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugen752-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAOBSIM
TTHto2B_M125	/TTHto2B_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAOBSIM
TTHtoNon2B_M125	/TTHtoNon2B_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAOBSIM
VBFHto2B_M125	/VBFHto2B_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAOBSIM
VBFHto2Tau_M125	/VBFHto2Tau_M125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAOBSIM
VBFHto2Tau_UnrelatedDecay_UnFiltered	/VBFHto2TauUnrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAOBSIM
VBFHto2Tau_UnrelatedDecay_Filtered	/VBFHto2TauUnrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAOBSIM
VBFHto2Wto2L2Nu_M125	/VBFHto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugen752-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAOBSIM
WminusH_Hto2B_WtoLNU	/WminusH_Hto2B_WtoLNU_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAOBSIM
WplusH_Hto2B_WtoLNU	/WplusH_Hto2B_WtoLNU_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAOBSIM
WminusHto2Tau_UnrelatedDecay_Filtered	/WminusHto2TauUnrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAOBSIM
WminusHto2Tau_UnrelatedDecay_UnFiltered	/WminusHto2TauUnrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAOBSIM
WplusHto2Tau_UnrelatedDecay_Filtered	/WplusHto2TauUnrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAOBSIM
WplusHto2Tau_UnrelatedDecay_UnFiltered	/WplusHto2TauUnrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAOBSIM
ZH_Hto2B_Zto2L	/ZH_Hto2B_Zto2L_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAOBSIM
ZH_Hto2B_Zto2Q	/ZH_Hto2B_Zto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAOBSIM
WminusH_Hto2B_Wto2Q	/WminusH_Hto2B_Wto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAOBSIM
WplusH_Hto2B_Wto2Q	/WplusH_Hto2B_Wto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAOBSIM
TbarBtoLminusNuB_s_channel_4FS	/TbarBtoLminusNuB-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAOBSIM
TBbartoplusNuBbar_s_channel_4FS	/TBbartoplusNuBbar-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAOBSIM
TbarWplusto2L2Nu	/TbarWplusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAOBSIM
TbarWplusto4Q	/TbarWplusto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAOBSIM
TbarWplustoLNU2Q	/TbarWplustoLNU2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAOBSIM
TWminusto2L2Nu	/TWminusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAOBSIM
TWminusto4Q	/TWminusto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAOBSIM
TWminustoLNU2Q	/TWminustoLNU2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAOBSIM
TTto2L2Nu	/TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAOBSIM
TTto4Q	/TTto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAOBSIM
TTtoLNU2Q	/TTtoLNU2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAOBSIM
TTWH	/TTWH_TuneCP5_13p6TeV_madgraph-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAOBSIM
TTWW	/TTWW_TuneCP5_13p6TeV_madgraph-madspin-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAOBSIM
TbarBQ_t_channel_4FS	/TbarBQ_t-channel_4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAOBSIM
TBbarQ_t_channel_4FS	/TBbarQ_t-channel_4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAOBSIM
TT	/TT_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAOBSIM
TTZH	/TTZH_TuneCP5_13p6TeV_madgraph-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAOBSIM
TTZZ	/TTZZ_TuneCP5_13p6TeV_madgraph-madspin-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAOBSIM
WtoLNU_1J_madgraphMLM	/WtoLNU-4Jets_1J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAOBSIM
WtoLNU_2J_amcatnloFXFX	/WtoLNU-2Jets_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAOBSIM
WtoLNU_2J_madgraphMLM	/WtoLNU-4Jets_2J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAOBSIM
WtoLNU_3J_madgraphMLM	/WtoLNU-4Jets_3J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAOBSIM
WtoLNU_4J_madgraphMLM	/WtoLNU-4Jets_4J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAOBSIM
WtoLNU_OJ_amcatnloFXFX	/WtoLNU-2Jets_OJ_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAOBSIM
WtoLNU_1J_amcatnloFXFX	/WtoLNU-2Jets_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAOBSIM
WtoLNU_amcatnloFXFX	/WtoLNU-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAOBSIM
WtoLNU_madgraphMLM	/WtoLNU-4Jets_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAOBSIM
WW	/WW_TuneCP5_13p6TeV_pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAOBSIM
WZ	/WZ_TuneCP5_13p6TeV_pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAOBSIM
ZZ	/ZZ_TuneCP5_13p6TeV_pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAOBSIM
WWto2L2Nu_powheg	/WWto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAOBSIM
WWto4Q_powheg	/WWto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAOBSIM
WWtoLNU2Q_powheg	/WWtoLNU2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOAOBSIM
WZto2L2Q_powheg	/WZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAOBSIM
WZto3LNU_powheg	/WZto3LNU_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOAOBSIM
	/WZto3LNU_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v2/NANOAOBSIM
	/WZto3LNU_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6_ext1-v2/NANOAOBSIM

Continued on next page

Table 1: NanoAOD 2023BPix datasets used in the analysis (continued)

Sample	NanoAOD dataset
WZtoLNu2Q_powheg	/WZtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOADSIM
ZZto2L2Nu_powheg	/ZZto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOADSIM
ZZto2L2Q_powheg	/ZZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOADSIM
ZZto2Nu2Q_powheg	/ZZto2Nu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOADSIM
ZZto4L_powheg	/ZZto4L_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOADSIM
WWW_4F	/WWW_4F_TuneCP5_13p6TeV_amcatnlo-madspin-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOADSIM
WWZ_4F	/WWZ_4F_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v3/NANOADSIM
WZZ	/WZZ_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOADSIM
ZZZ	/ZZZ_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v2-v2/NANOADSIM
ZHto2Tau_UncorrelatedDecay_Filtered	/ZHto2TauUncorrelatedDecay_Filtered_M-125_CP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v5/NANOADSIM
ZHto2Tau_UncorrelatedDecay_UnFiltered	/ZHto2TauUncorrelatedDecay_M-125_CP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23BPixNanoAODv12-130X_mcRun3_2023_realistic_postBPix_v6-v5/NANOADSIM